

Analysis of bird point counts

Report

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1 Introduction

Anthropogenic climate change could influence migratory birds' behavior. Conventionally, warmer temperatures would change when plants and insects appear, which could then shift bird migration timing. This research is interested in studying a population of barn swallows (*Hirundo rustica*) on Lopez Island. Barn swallows are migratory birds that migrate from North American breeding grounds to wintering areas in Central and South America during the late summer. They are semi-colonial, often nesting alongside one another in larger man-made structures.

The question of interest is: how do barn swallow abundance on Lopez Island change during spring migration with respect to habitat characteristics and year?

I will analyze two parts. First I will determine whether the presence of barns and trees influences barn swallow abundance, then I will investigate whether migration timing has changed over time by testing whether the seasonal abundance pattern varies among years.

2 Methods

2.1 Data

Beginning in 2001, researchers established 14 survey sites near roosting and feeding habitats. Because of budget limits, surveys were done biennially and during the spring northward migration only. Each survey season includes 12 weekly surveys and uses 5-minute point counts at each site. Two technicians conduct the surveys. Before each survey, they flip a coin to decide which technician surveys which set of 7 locations. At each location, the technician records the number of barn swallows seen within a 50-meter radius.

The full data contains these variables:

- year: calendar year

- week: week of the survey; the first survey begins in mid-February
- site: ID of the point count location
- lon: relative longitude of the point count location
- lat: relative latitude of the point count location
- trees: binary indicator of tree presence/absence (0/1) within the survey circle
- barns: binary indicator of barn presence/absence (0/1) within the survey circle
- tech_ID: initials of the technician surveying a given location
- count: number of barn swallows counted in 5 minutes

2.2 Modeling

2.2.1 initial model

First I will test a Poisson Model. The three components are: Distribution of the data:

$$Y_i \sim \text{Poisson}(\mu_i)$$

where μ_i the expected barn swallow count for observation i

Link function:

$$\log(\mu_i) = \eta_i$$

Linear predictor:

$$\eta_i = \beta_0 + \beta X$$

```
##           Estimate Std. Error  z value  Pr(>|z|)
## (Intercept)  0.2188263  0.0367299   5.9577 2.558e-09
## barns1       0.8979801  0.0308698  29.0893 < 2.2e-16
## trees1      -0.3547244  0.0301429 -11.7681 < 2.2e-16
## week        0.1084648  0.0041334  26.2408 < 2.2e-16
##
## n = 1848 p = 4
## Deviance = 2134.35712 Null Deviance = 3656.67779 (Difference = 1522.32068)
```

All model coefficients are “significant” at the $\alpha = 0.05$ level.

2.2.2 goodness of fit

```
## [1] 0.1732085
```

The p-value is pretty big, so we fail to reject H_0 . There is no evidence of lack of fit for the Poisson model.

2.2.3 check for overdispersion

```
## [1] 1.030962
```

This estimate of dispersion is close to 1 so there was no indication of substantial overdispersion in the initial model.

2.2.4 Model diagnostic:

The Pearson residuals vary systematically across sites in the initial model. This indicates that the sites are not independent. Thus we need to include random effect for site and use a Poisson GLMM.

2.2.5 Part A: Habitat characteristics effect

Do barns and/or trees affect barn swallow abundance?

I compare four candidate models while using week and year as control variables to account for the fact that counts vary seasonally and over time: 1: neither barns nor trees 2: barns only 3: trees only 4: barns + trees (full habitat model)

Model comparison: calculating AIC

AIC values for Question A models

```
## A1 (no habitat):      AIC = 6806.07, deltaAIC = 41.86
## A2 (barns only):      AIC = 6791.57, deltaAIC = 27.36
## A3 (trees only):      AIC = 6805.36, deltaAIC = 41.15
```

```
## A4 (barns + trees, linear week):    AIC = 6764.21, deltaAIC = 0.00
```

From the result, A4(model with both barns and trees terms) has deltaAIC=0 and is the best model.

In addition, I want to test whether the week effect is linear. Fit a new model with quadratic week term:

```
## A5 (barns + trees, quadratic week): AIC = 6737.52, deltaAIC = 0.00
```

From the result, A5(with the quadratic week term) is a better model than A4.

The model that includes the quadratic week effect received substantially stronger support from AIC than the model with only a linear week effect (deltaAIC = 26.69), indicating that the nonlinear seasonal pattern was important to account for.

2.2.6 Part B: Year effect on migration timing

Has the seasonal peak of barn swallow abundance shifted over years?

I compare two candidate models while using barns and trees as control variables and retained the quadratic week term:

1: week + week² + year + habitat controls (no interaction) ->this is A5
2: (week+week²) * year + habitat controls (interaction)

Model selection: calculating AIC

```
## B1 (no interaction): AIC = 6737.52, deltaAIC = 0.00
```

```
## B2 (interaction):    AIC = 6738.36, deltaAIC = 0.84
```

From the result, B1(no week year interaction) is probably a better model, although these two do not have significant difference.

```
#Likelihood ratio test
```

```
# Direct test of whether the week:year term improves the model
```

```
anova(B1, B2, test = "LRT")
```

```
## Data: data
```

```
## Models:
```

```
## B1: count ~ barns + trees + week_s + I(week_s^2) + year_s + (1 | site)
```

```
## B2: count ~ barns + trees + (week_s + I(week_s^2)) * year_s + (1 | site)
```

```
##      npar    AIC    BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
```

```
## B1      7 6737.5 6776.2 -3361.8    6723.5
```

```
## B2      9 6738.4 6788.1 -3360.2    6720.4 3.1633  2    0.2056
```

The p-value is big, we fail to reject H₀, so there is no evidence that the interaction improves the model, which means that there is no evidence that seasonal peak of barn swallow abundance shifted over years.

2.2.7 Assumption checking for B1

Overdispersion

```
## [1] 1.016567
```

ϕ is close to 1, so there is no substantial overdispersion in this model.

check for zeros

```
## Observed proportion of zeros: 0.145
## Expected under Poisson: 0.056
# see where zeros are distributed
weeks <- sort(unique(data$week))
prop_zero_by_week <- sapply(weeks, function(w) {
  round(mean(data$count[data$week == w] == 0), 3)
})

zero_by_week <- data.frame(week = weeks, prop_zero = prop_zero_by_week)
print(zero_by_week)

##      week prop_zero
## 1      1      0.519
## 2      2      0.377
## 3      3      0.279
## 4      4      0.162
## 5      5      0.084
## 6      6      0.117
## 7      7      0.045
## 8      8      0.071
## 9      9      0.065
## 10     10      0.019
## 11     11      0.000
## 12     12      0.000

cat(sprintf("Week with highest zero proportion: week %d (0.1f%%)\n",
            weeks[which.max(prop_zero_by_week)],
            max(prop_zero_by_week) * 100))

## Week with highest zero proportion: week 1 (51.9%)

cat(sprintf("Week with lowest zero proportion: week %d (0.1f%%)\n",
            weeks[which.min(prop_zero_by_week)],
            min(prop_zero_by_week) * 100))

## Week with lowest zero proportion: week 11 (0.0%)
```

The higher-than-expected proportion of zero counts reflects structural zeros in early survey weeks, when barn swallows had not yet arrived at the study site.

Thus, the final habitat model was a Poisson generalized linear mixed model (GLMM) with a log link and a random intercept for site.

$$Y_i \sim \text{Poisson}(\mu_i)$$

$$\log(\mu_i) = \eta_i$$

$$\eta_i = \beta_0 + \beta_1 \text{barns}_i + \beta_2 \text{trees}_i + \beta_3 \text{week}_{s,i} + \beta_4 \text{week}_{s,i}^2 + \beta_5 \text{year}_{s,i} + b_{\text{site}(i)}$$

where

$$b_{\text{site}(i)} \sim N(0, \sigma_{\text{site}}^2)$$

and μ_i is the expected barn swallow count for observation i .

3 Results

This is the finalized coefficient table:

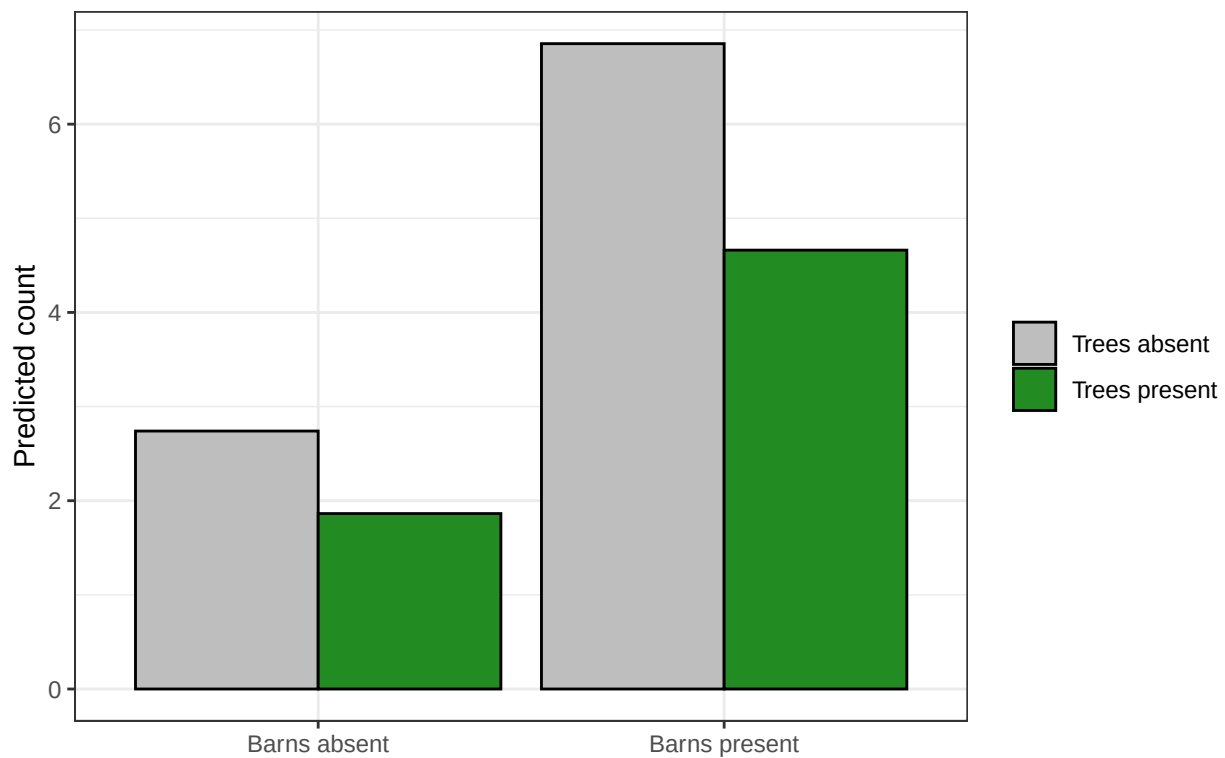
##	term	exp_est	exp_lower	exp_upper
## 1	Intercept	2.740	2.585	2.905
## 2	Barns present	2.502	2.261	2.768
## 3	Trees present	0.680	0.624	0.742
## 4	Week (linear)	1.494	1.449	1.541
## 5	Week (quadratic)	0.916	0.887	0.946
## 6	Year	1.005	0.979	1.033

Sites with barns had about 2.5 times higher predicted counts than sites without barns (CI: 2.261–2.768). Sites with trees had predicted counts that were 32% lower than sites without trees (multiplicative effect = 0.68, 95% CI: 0.624–0.742). The effect of year on abundance was negligible (multiplicative effect = 1.005, 95% CI: 0.979–1.033), providing no evidence of a long-term trend over the study period.

3.1 part A prediction plot

Predicted barn swallow abundance by habitat

Predictions at mean week and mean year



3.2 Part B prediction plot

Shows whether the peak shifts left (earlier) or right (later) over time

Shows whether the peak shifts left (earlier) or right (later) over time

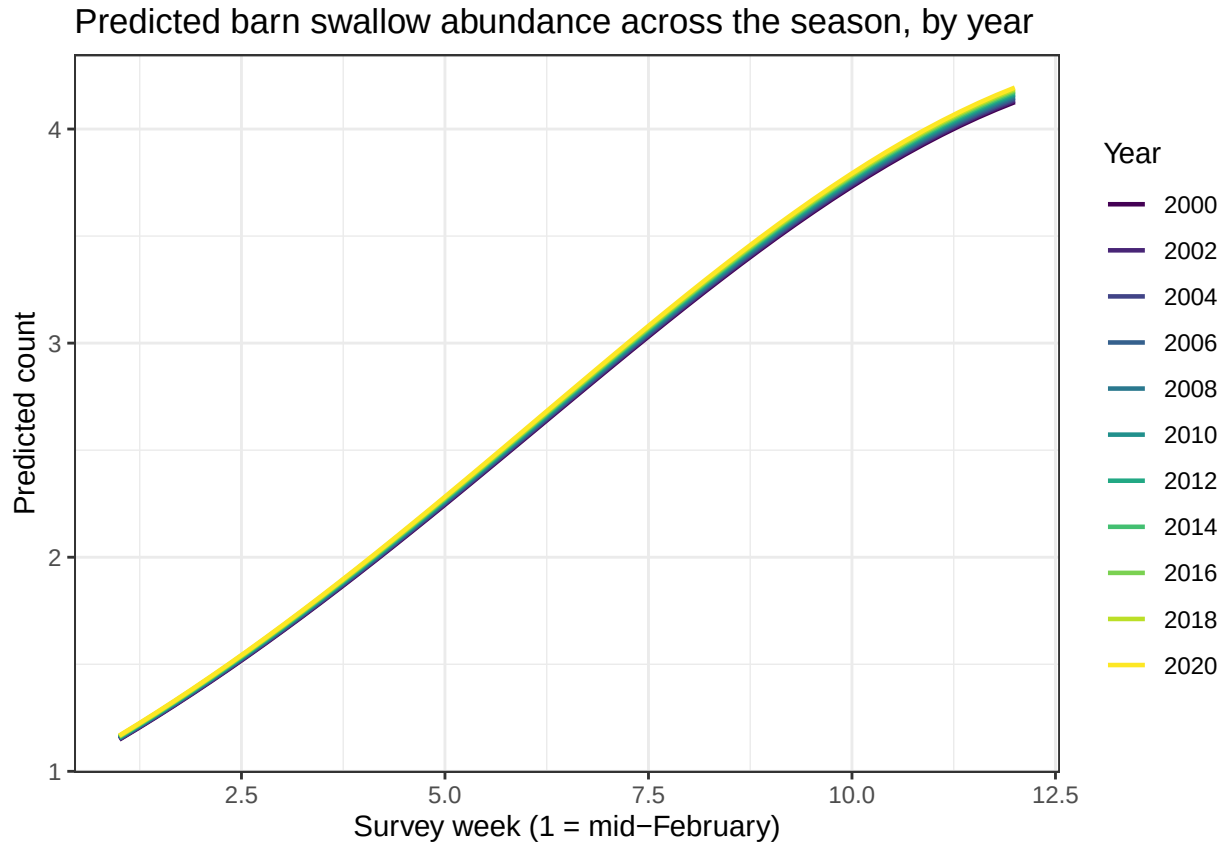
```
weeks_seq_s <- seq(min(data$week_s), max(data$week_s), length.out = 50)
```

```
pred_B <- expand.grid(  
  week_s = weeks_seq_s,  
  year_s = unique(data$year_s),  
  barns = levels(data$barns)[1],  
  trees = levels(data$trees)[1]  
)
```

```
pred_B <- within(pred_B, {  
  week_orig <- week_s * sd(data$week) + mean(data$week)  
  year_orig <- round(year_s * sd(data$year) + mean(data$year))  
  fit <- predict(B1, newdata = pred_B,  
                 type = "response", re.form = NA)  
})
```



```
ggplot(pred_B, aes(x = week_orig, y = fit,
                   color = factor(year_orig), group = factor(year_orig))) +
  geom_line(linewidth = 0.8) +
  scale_color_viridis_d(name = "Year") +
  labs(title = "Predicted barn swallow abundance across the season, by year",
       x = "Survey week (1 = mid-February)", y = "Predicted count") +
  theme_bw()
```



4 Discussion

Barn swallow abundance was strongly associated with habitat characteristics. Models including both barn and tree effects received the greatest support from AIC. As shown in the first figure, counts were the highest when sites contain barns but no trees. Sites containing barns generally have larger counts than sites that don't.

There was little evidence that migration timing changed across years. As shown in the second figure, the predicted seasonal curves were nearly identical across years. The model including interactions between year and seasonal abundance patterns was not supported relative to the simpler model, and the likelihood ratio test was not significant.

Therefore, while habitat characteristics influenced barn swallow abundance, the analysis did not provide evidence that the timing of seasonal abundance patterns shifted over the study period. However, the biennial survey design and the restriction to the pre-peak migration period may have limited the power to detect subtle temporal shifts.

5 Reference

Data provided by Professor Scheuerell